The Rules

6.0.1
Video - Devices in a Bubble

Click Play in the figure to view a video explaining how a network device operates within a network.

6.0.2
Communications Fundamentals

Networks vary in size, shape, and function. They can be as complex as devices connected across the Internet, or as simple as two computers directly connected to one another with a single cable, and anything in between. However, simply having a wired or wireless physical connection between end devices is not enough to enable communication. For communication to occur, devices must know "how" to communicate.

People exchange ideas using many different communication methods. However, all communication methods have the following three elements in common:

- **Message source (sender)** – Message sources are people, or electronic devices, that need to send a message to other individuals or devices.
- **Message destination (receiver)** – The destination receives the message and interprets it.
- **Channel** – This consists of the media that provides the pathway over which the message travels from source to destination.

6.0.3
Communication Protocols

Sending a message, whether by face-to-face communication or over a network, is governed by rules called protocols. These protocols are specific to the type of communication method being used. In our day-to-day personal communication, the rules we use to communicate over one medium, like a telephone call, are not necessarily the same as the rules for using another medium, such as sending a letter.

The protocol of sending a letter is similar to communication that occurs in computer networks.

Click each button for an analogy and a network example of the communication process.

Analogy | Network

Analogy

Click Play in the figure to view an animation of two people communicating face-to-face.

Prior to communicating, they must agree on how to communicate. If the communication is to be face-to-face, they must first see each other. Next, when they have a message to share, they must be able to format that message in a way that is understandable.

If someone uses the English language, but poor sentence structure, the message can easily be misunderstood. Each of these tasks describes protocols that are used to accomplish communication.

6.0.4
Rule Establishment

Before communicating with one another, individuals must use established rules or agreements to govern the conversation. Consider this message for example:

```
Humans communicating between greater metro. Or is very difficult to understand messages that are not correctly formatted and about follow the established rules and protocols. A structure is grammatic, be tongue, the punctuation and the sentence make the configuration header compressed get mention individual information.
```

Notice how it is difficult to read the message because it is not formatted properly. It should be written using rules (i.e., protocols) that are necessary for effective communication. The example shows the message which is now properly formatted for language and grammar.

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Rules govern communication between humans. It is very difficult to understand messages that are not correctly formatted and do not follow the established rules and protocols. The structure of the grammar, the language, the punctuation and the sentence make the configuration header understandable for many different individuals.
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Protocols must account for the following requirements to successfully deliver a message that is understood by the receiver:

- An identified sender and receiver
- Common language and grammar
- Agreed upon method of delivery
- Confirmation or acknowledgment requirements

6.0.5
Network Protocol Requirements

The protocols that are used in network communications share many of these fundamental traits. In addition to identifying the source and destination, computer and network protocols define the details of how a message is transmitted across a network. Common computer protocols include the following requirements:

- Message encoding
- Message formatting and encapsulation
- Message size
- Message timing
- Message delivery options

6.0.6
Message Encoding

One of the first steps in sending a message is encoding. Encoding is the process of converting information into another acceptable form, for transmission. Decoding reverses this process to interpret the information.

Click each button for an analogy and a network example of message encoding.

Analogy | Network

Analogy

Imagine a person calls a friend to discuss the details of a beautiful sunset. Click Play in the figure to view an animation of message encoding.

To communicate the message, she converts her thoughts into an agreed upon language. She then speaks the words using the sounds and inflections of spoken English. The friend receives the message, hears the sounds, translates and decodes the sounds to understand the message he received.

6.0.7
Message Formatting and Encapsulation

When a message is sent from source to destination, it must use a specific format or structure. Message formats depend on the type of message and the channel that is used to deliver the message.

Click each button for an analogy and a network example of message formatting and encapsulation.

Analogy | Network

Analogy

A common example of requiring the correct format in human communications is when sending a letter. Click Play in the figure to view an animation of formatting and encapsulating a letter.

The letter has the address of the sender and receiver, each located at the proper place on the envelope. If the destination address and formatting are not correct, the letter is not delivered.

The process of placing one message format (the letter) inside another message format (the envelope) is called encapsulation. De-encapsulation occurs when the process is reversed by the recipient and the letter is removed from the envelope.

6.0.8
Message Size

Another rule of communication is message size.

Click each button for an analogy and a network example of message size.

Analogy | Network

Network

Likewise, when a long message is sent from one host to another over a network, it is necessary to break the message into smaller pieces, as shown in Figure 2. The rules for how large these pieces or blocks of communication are transmitted across the network are very strict. They can also be different, depending on the channel used. Frames that are too long or too short are not delivered.

The size restrictions of frames require the source host to break a long message into individual pieces that meet both the minimum and maximum size requirements. The long message will be sent in separate frames, with each frame containing a piece of the original message. Each frame will also have its own addressing information. At the receiving host, the individual pieces of the message are reconstructed into the original message.

6.0.9
Message Timing

Message timing is also very important in network communications. Message timing includes the following:

- **Flow Control** – This is the process of managing the rate of data transmission. Flow control defines how much information can be sent and the speed at which it can be delivered. For example, if one person speaks too quickly, it may be difficult for the receiver to hear and understand the message. In network communications, there are many factors that can slow down the flow of information as it is processed by network devices.
- **Response Timeout** – If a person asks a question and does not hear a response within an acceptable amount of time, the person assumes that no answer is coming back. The person can decide that this lack of response is due to something that is not right. The person can then decide how long to wait for a response and what action to take if a response never occurs.
- **Access Method** – This determines when someone can send a message. For example, in a face-to-face chat, it is fine to use an animation of two people talking at the same time, then a business or international discussion, and it is necessary for the rule to be kept off and back again. Likewise, when a device wants to transmit on a wireless LAN, it is necessary for the WLAN network interface card (NIC) to determine whether the wireless medium is available.

6.0.10
Message Delivery Options

A message can be delivered in different ways.

Click each button for an analogy and a network example of message delivery options.

Analogy | Network

Analogy

Sometimes, a person wants to communicate information to a single individual. At other times, the person may need to send information to a group of people at the same time, or even to all people in the same area.

Click the unicast, multicast, and broadcast buttons in the figure for an example of each.

6.0.11
A Note About the Node Icon

Networking documents and topologies often represent networking and end devices using a node icon. Nodes are typically represented as a circle. The figure shows a comparison of the three different delivery options using node icons instead of computer icons.

6.0.12
Check Your Understanding - The Rules

Check your understanding of the rules of successful communication by choosing the BEST answer to the following questions.

1. What is the process of converting information into the proper form for transmission?
   - Formatting
   - Encoding
   - Encapsulation

2. Which step of the communication process is concerned with properly identifying the address of the sender and receiver?
   - Formatting
   - Encoding
   - Encapsulation

3. Which three are components of message timing? (Choose three.)
   - Flow control
   - Sequence numbers
   - Access method
   - Acknowledgment time
   - Response timeout

4. Which delivery method is used to transmit information to one or more end devices, but not all devices on the network?
   - Unicast
   - Multicast
   - Broadcast

Check | Show Me | Reset